INCH-POUND

MIL-C-24643/38C
14 March 1994
SUPERSEDING
MIL-C-24643/38B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSECM (INCLUDING VARIATION LSECMA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSECM (unarmored) and LSECMA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, in accordance with ASTM B 286, size 20-7, tin coated. Maximum diameter 0.039 inch.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Crosslinked polyethylene insulation. Nominal wall thickness 0.023 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.

Fourth - Two conductors (one black and one white) cabled together with a nominal lay of 2.5 inches to form each a pair.

Fifth - Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.

Sixth - Braided shield of tin coated copper. Minimum coverage of 85 percent

* Seventh - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

AMSC N/A FSC 6145

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Construction (watertight) (continued)

Groups of 7 (8 groups)

First - Copper conductor, AWG no. 18, class B stranding, (7 by 0.0152 inch) uncoated. Maximum diameter 0.050 inch.

Second - Ethylene propylene rubber or cross-linked polyethylene insulation, white. Nominal wall thickness 0.008 inch. Colored insulation five black and two white conductors for each group.

Third - Seven conductors, five black and two white, with the white conductors adjacent in the layer of six, cabled with a lay not grater than 2 inches, to form each group.

* Fourth - Two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

ASSEMBLY:

* First - Five pairs cable to form a core. Three pairs and eight groups of seven cabled over the core, with a lay not greater than 24 times the pitch diameter of the layer. Pair nos. 1 through 5 in core; the sequence of the outer layer - pair no. 6, groups nos. 1, 2 and 3; pair no. 7; groups nos. 4, 5 and 6; pair no. 8; group nos. 7 and 8. Fillers shall be used as necessary to form a well rounded assembly.

Second - Binder tape applied helically with overlap.

Construction variant for type LSECM

* Third - Cross-linked polyolefin jacket. Cable jack thickness 0.075 inch minimum average.

Construction variant for type LSECMA

* Third - Cross-linked polyolefin jacket. Cable jacket thickness 0.075 inch minimum average.

* Fourth - Braided metal armor.

NOTE: 1. Variant LSECM completed cable diameters: Minimum 1.290 inches, maximum 1.370 inches. Military part number M24643/38-01UN.

2. Variant LSECMA completed cable diameters: Minimum 1.340 inches, maximum 1.420 inches. Military part number M24643/38-02AN.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	
ASTM B 286, size 20-7	11.03
AWG no 18 Class B	7.47
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Shield to shield	500
Insulation resistance - Megohms/1000 feet minimum	
* Conductor to shield	500
* Component shield to shield	100
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Capacitance	
Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum	30
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75

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EXAMINATION AND TESTS (continued):	Requirements
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - (type LSECM only) cycles, minimum	125
* Cable sealant removability	No failure
Armor (type LSECM only) - conformance to material, construction and coverage	No failure
* Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

	Requirements
* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:	
Aging and compatibility (cable) (125± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield insulation group covering	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45

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QUALIFICATION INSPECTION:

	Requirements
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:
1000 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project No. 6145-2057-38)

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